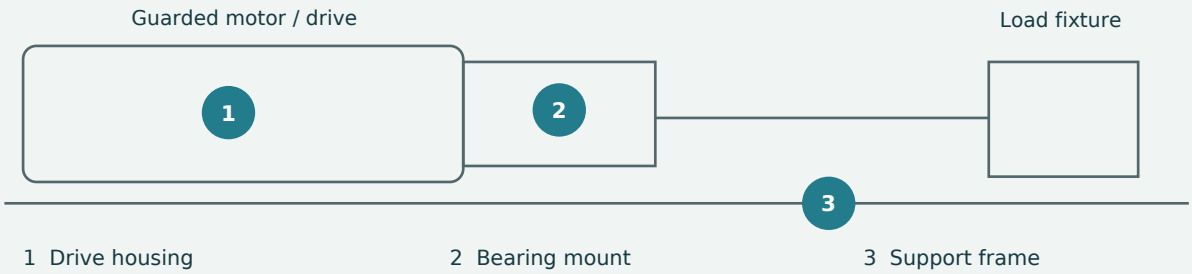
CONFLUENCE-01 v0.1

A synthetic sense of changing physical activity.

Design goal: turn the changing relationship between heat flow, magnetic fields and structural vibration into one stable, learnable percept. First a machine-readable map; later a visual or tactile form a person could learn to use.

Example: a guarded workshop motor stops. Its vibration and magnetic pattern change quickly, while heat continues to leave the housing. The system would represent thermal after-effect rather than confuse lingering heat with continued mechanical operation. A new conflicting pattern would remain unfamiliar change, not a guessed fault.



Illustrative layout only. Registered sensor zones; no interpolation into unsensed space.

Proposed state	Meaning	Candidate tactile vocabulary
Quiet	No substantial activity pattern in the measured bands; not an all-clear.	One short tap on request.
Familiar operation	History fits a commissioned operating pattern.	A regular, restrained pulse.
Thermal after-effect	Slow heat response persists after fast activity subsides.	Paired, slow taps.
Unfamiliar change	Valid data differs from learned patterns. Cause unresolved.	A distinct three-tap sequence.
Unavailable	Required sensor, contact, timing or model coverage is inadequate.	A separate long-double cue plus explicit visual status.

Only one selected zone is felt at a time through a remote piezoelectric tactor. A small map shows all sampled zones with shape and text as well as color. The tactile code is a proposal to test, not an established language or training result. Silence never means safe.

Maturity: concept and experiment plan. No prototype, dataset, new biological modality, perceptual benefit or novelty has been demonstrated.

Three registered zones, three real observables

Choose fixed sensing points on one guarded, isolated low-voltage motor/drive assembly. Compare this with a mobile scanner: moving pose, air gaps and hand motion make repeatable magnetic, vibration and direct heat-flux measurements harder. Room-scale scanning is deferred. The fixed assembly is Asteris' design choice, not a user-imposed constraint.

Input	Hardware and physical meaning	Commissioning requirement
Heat flow	A thin thermopile patch measures normal heat-flux density q_n THROUGH its own surface in W/m^2. Local surface temperature provides context [2,4].	Measure sign, scale, thermal response and installation effects. The patch perturbs heat transfer; it is not automatically the undisturbed surface flux or a 3D flow vector.
Magnetic field	A fixed, offset three-axis sensor measures B in microtesla or millitesla. Select range and usable band from measured fields; TI Hall sensors are one candidate family [5].	Keep motor self-field when it is the signal of interest. Check background, saturation, drift, wiring and mount material. Do not use magnetic mounting clamps.
Structural vibration	A rigidly mounted triaxial accelerometer measures acceleration in m/s^2; extract calibrated band energy and time features. Industrial development kits are available [6].	Measure mounting transfer function, clipping, noise and antialias behavior. Acceleration does not by itself measure stress, strain or damage.
Geometry / acquisition	Each zone is a registered triad: rigid accelerometer attachment, nearby thermal patch and offset magnetic mount. A common logger captures original samples and hardware timestamps.	One enclosure is not optimal for all three sensors. Record geometry, contact method and changes. Keep processing electronics and the tactile actuator away from measurement locations.

One representation; different response times

$$\text{state}_i(t) = f_{\text{frozen}}(q_{n,i}\ \text{history}, B_i\ \text{history}, a_i\ \text{history}, \text{validity}, \text{age})$$

Retain physical units, raw data and per-channel uncertainty. Model explicit lags and commissioning baselines; do not add quantities with different units. Start with a simple state classifier and lag features. A graph neural network is unnecessary for this first experiment.

Magnetic and vibration features can change much faster than heat. Set sample rates and synchronization tolerances from actual sensor bandwidths, the chosen phenomenon and measured delay. A fast display refresh cannot manufacture fresh thermal information. Distinguish stale samples from a real slow response, valid-but-unfamiliar data from hardware failure, and learning from online normalization that could erase a change.

The map covers the three instrumented zones only. Cross-channel relationships are evidence to investigate, not a proof of cause or an image of all energy moving through the object.

Separate sensing value from interface value

Stage	Proposed experiment and decision
Calibrate the assembly	Characterize a reference vibration input, controlled magnetic input and a heat-flux installation. Compare repeated remounts, contact pressure/interface changes, thermal boundary conditions and temperature references. Test electronics and tactor interference with matched on/off trials.
Collect controlled episodes	Record normal speed/load changes within equipment ratings, stop/cooling, separately applied heater/field/shaker disturbances, mixed confounders, dropouts and remounts. Preserve independent ground truth; do not give commands or answer labels to the model as hidden inputs.
Test physical information	Compare each single modality, all pairs, all three and a vibration-plus-surface-temperature alternative. Choose baselines on validation data. Use equal tuning budgets, frozen thresholds and held-out whole sessions, remounts and - when available - separate units.
Measure honest performance	Report state error, abstention/coverage, false events per valid hour, latency, calibration and resource burden. Compare at matched coverage or show the full error-versus-abstention tradeoff. Thermal delays require state-specific timing criteria.
Test the learned percept	Counterbalance compact glyphs against separate readings using the same underlying features, predictions, quality information and access. Test tactile encoding separately with matched state information. Use repeated learning sessions, transfer trials, secondary-task burden and comfort. Freeze the vocabulary during evaluation.
Accept or simplify	After a pilot, pre-register a practically useful improvement target and uncertainty analysis. If magnetic or flux sensing adds no held-out value at its cost and installation burden, drop it. If integrated encoding adds no usable performance or learning benefit, retain ordinary monitoring and withdraw the new-sense framing.

Two ways the proposal could fail

False thermal meaning: the patch's contact or mounting history could dominate its signal. A classifier might learn the installation rather than the machine. Repeatable remount and surface-temperature comparisons are therefore decisive.

False perceptual novelty: five learned buzzes may function only as alerts. Faster recognition or subjective familiarity does not prove a new biological sense. Useful transfer, retention and performance under a competing task would support the narrower sensory-augmentation claim.

The first system is advisory. Existing equipment protections remain responsible for stopping machinery. Identical measured histories can have different causes; the system must permit ambiguity rather than promise universal diagnosis.

What the Court added

Reviewer	Retained contribution / correction
Grok	Keep fixed registered nodes and test added information against simpler sensors. Strongest remaining objection: the heat patch can create misleading lag patterns. Add a dedicated remount/contact-perturbation study.
Gemini	Use explicit time history: lingering heat after shutdown can be a meaningful state. Replaced unsupported instantaneous vectors, stress inference, arbitrary thresholds and a forced graph network with a simpler representation.
Meta	Separate signal validity, sample age and unfamiliar activity. Retain self-field and startup transients as possible information. Use one selected-zone tactile state; evaluate interface benefits separately from the classifier.
Asteris / ChatGPT	Selected the bounded three-zone design and five-state target; checked primary precedents and retained physical limits. Court convergence followed a shared brief and corrective round, not independent experiments or proof.

Existing work and the actual research question

ADI OtoSense already combines vibration, temperature and magnetic sensing [1]. Sensory augmentation also has experimental precedents: the 2016 feelSpace study reported training-related and subjective perceptual changes, while its complex navigation performance improvement was not significant [3]. Neither combining sensors nor making a tactile cue is claimed as our invention.

The open question is whether a stable, time-aware representation of these measurements helps machines and people distinguish changing activity more reliably, with less attention, than well-designed simpler alternatives. Direct heat flux must earn its place against surface temperature.

[1] Analog Devices, OtoSense Smart Motor Sensor Analytics Guide
Primary manufacturer guide; sensor-combination precedent. HTML version timed out; PDF opened.

[2] Hukseflux, HFP01 / HFP03 user manual
Primary measurement principle and installation limitations. Not a selection of this building/soil plate for a motor.

[3] Konig et al., Learning New Sensorimotor Contingencies, PLOS ONE, 2016
Primary sensory-augmentation study; benefit and limitations both retained.

[4] greenteg, gSKIN-XM heat-flux sensor
Example thin sensor available for controlled applications; mounting qualification remains open.

[5] Texas Instruments, TMAG5170 product documentation
Example three-axis Hall-sensor family; achievable installed noise and bandwidth are not asserted.

[6] STMicroelectronics, STEVAL-STWINBX1 / STWIN.box
Example industrial sensing development platform, not an integrated heat-flux solution.

Open before procurement: assembly and operating envelope; target task and band; exact sensors/mounts; thermal calibration; lag windows; minimal useful benefit; training/sample size; component and installation cost. No purchases or hardware tests performed.

Credits: Anubis - design direction. Asteris Solenne / ChatGPT - proposal, source audit and consolidation. Grok, Gemini and Meta - live design critiques.